Lesson 34: Mixed Strategies, Nash Existence, and the Minimax Theorem

AI/ML Mathematics

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1 Mixed Strategies and Mixed Extensions

1.1 Definitions

In Lesson 33 we saw that some games (e.g., Matching Pennies) have no pure strategy Nash equilibrium. To guarantee existence, we enlarge each player's strategy space from finitely many pure strategies to the simplex of probability distributions over them.

Definition 1.1 (Mixed Strategy)**.** Let $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ be a finite game with $S_i = \{s_i^1, \dots, s_i^{m_i}\}$ for player i. A *mixed strategy* for player i is a probability distribution $\sigma_i \in \Delta(S_i)$, where

$$\Delta(S_i) = \left\{ \sigma_i \in \mathbb{R}^{m_i} : \sigma_i(s_i^j) \geq 0 \text{ for all } j, \ \sum_{j=1}^{m_i} \sigma_i(s_i^j) = 1 \right\}$$

is the $(m_i - 1)$-simplex over S_i. A *mixed strategy profile* is $\sigma = (\sigma_1, \dots, \sigma_n) \in \Delta := \prod_{i\in N} \Delta(S_i)$.

Definition 1.2 (Support)**.** The *support* of a mixed strategy σ_i is $\operatorname{supp}(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$. A pure strategy s_i is identified with the degenerate distribution placing probability 1 on s_i.

Definition 1.3 (Mixed Extension)**.** The *mixed extension* of a finite game $\Gamma = (N, (S_i), (u_i))$ is the game $\bar{\Gamma} = (N, (\Delta(S_i)), (\bar{u}_i))$ where the expected payoff under independent mixing is

$$\bar{u}_i(\sigma) = \sum_{s\in S} \left(\prod_{j\in N} \sigma_j(s_j) \right) u_i(s) = \mathbb{E}_{s\sim\sigma}[u_i(s)].$$

We drop the bar and write $u_i(\sigma)$ when the context is clear.

Remark 1.4 (Multilinearity). The expected payoff $u_i(\sigma)$ is *multilinear*: linear in each σ_i when the other players' strategies are held fixed. Explicitly,

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i\in S_i} \sigma_i(s_i) \underbrace{\sum_{s_{-i}\in S_{-i}} \left(\prod_{j\neq i} \sigma_j(s_j) \right) u_i(s_i, s_{-i})}_{=:\, u_i(s_i, \sigma_{-i})}.$$

The quantity $u_i(s_i, \sigma_{-i})$ is the expected payoff to player i from the pure strategy s_i when opponents play σ_{-i}.

1.2 Mixed Nash Equilibrium and the Indifference Principle

Definition 1.5 (Mixed Strategy Nash Equilibrium)**.** A mixed strategy profile σ^* is a *Nash equilibrium* if

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*) \quad \text{for all } \sigma_i \in \Delta(S_i), \text{ for all } i \in N.$$

Key Result

Theorem 1.6 (Indifference Principle)**.** *Let σ^* be a mixed strategy Nash equilibrium. For each player i:*

(a) If $s_i, s_i' \in \operatorname{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) = u_i(s_i', \sigma_{-i}^*)$.

(b) If $s_i \in \operatorname{supp}(\sigma_i^*)$ and $t_i \notin \operatorname{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) \geq u_i(t_i, \sigma_{-i}^*)$.

That is, every pure strategy in the support yields the same expected payoff, and this common payoff is at least as large as the payoff from any strategy outside the support.

Proof. By multilinearity (Remark 1.4),

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s_i \in S_i} \sigma_i^*(s_i)\, u_i(s_i, \sigma_{-i}^*).$$

Let $v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$ denote the equilibrium payoff for player i.

Part (b): Suppose for contradiction that some $t_i \in S_i$ satisfies $u_i(t_i, \sigma_{-i}^*) > v_i^*$. Then the pure strategy t_i (viewed as a degenerate mixed strategy) gives payoff $u_i(t_i, \sigma_{-i}^*) > v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$, contradicting the NE condition. Hence $u_i(s_i, \sigma_{-i}^*) \le v_i^*$ for all $s_i \in S_i$.

Now v_i^* is a convex combination of the payoffs $u_i(s_i, \sigma_{-i}^*)$ with weights $\sigma_i^*(s_i) \ge 0$ summing to 1. Since every term in the combination satisfies $u_i(s_i, \sigma_{-i}^*) \le v_i^*$, if any term in the support were strictly less than v_i^*, some other term would need to be strictly greater (to make the average equal v_i^*), which we just ruled out. Therefore $u_i(s_i, \sigma_{-i}^*) = v_i^*$ for all $s_i \in \operatorname{supp}(\sigma_i^*)$, and $u_i(t_i, \sigma_{-i}^*) \le v_i^*$ for all $t_i \notin \operatorname{supp}(\sigma_i^*)$.

Part (a): Immediate from the equality $u_i(s_i, \sigma_{-i}^*) = v_i^*$ for all $s_i \in \operatorname{supp}(\sigma_i^*)$. □

Example 1.7 (Mixed NE in Matching Pennies)**.** In Matching Pennies, let player 1 play H with probability p and player 2 play H with probability q. By the indifference principle, player 1 is indifferent when:

$$u_1(H, q) = u_1(T, q) \implies q - (1-q) = -(q) + (1-q) \implies 2q - 1 = 1 - 2q \implies q = \tfrac{1}{2}.$$

Similarly, player 2 is indifferent when $p = 1/2$. The unique mixed NE is $\sigma_1 = (1/2, 1/2)$, $\sigma_2 = (1/2, 1/2)$, with expected payoffs $(0, 0)$.

Example 1.8 (Mixed NE in Battle of the Sexes)**.** In the Battle of the Sexes, let player 1 play O with probability p and player 2 play O with probability q. By the indifference principle:

$$\begin{aligned} u_1(O, q) = u_1(F, q) &\implies 3q = 1 - q \implies q = \tfrac{1}{4}, \\ u_2(p, O) = u_2(p, F) &\implies p = 3(1-p) \implies p = \tfrac{3}{4}. \end{aligned}$$

The mixed NE is $\sigma_1 = (3/4, 1/4)$, $\sigma_2 = (1/4, 3/4)$, with expected payoffs $(3/4, 3/4)$. Both players fare worse than in either pure NE.

RL/ML Connection

In RL, a stochastic policy $\pi(a|s)$ is precisely a mixed strategy in the stage game at state s. The indifference principle has a direct analog: in an optimal stochastic policy, every action in the support must yield the same Q-value. If some action had a strictly higher Q-value, the agent should place all probability on it.

2 Nash's Existence Theorem

The central result of this lesson guarantees that every finite game has at least one Nash equilibrium in mixed strategies. The proof uses Kakutani's fixed point theorem.

2.1 Kakutani's Fixed Point Theorem

Definition 2.1 (Upper Hemicontinuity)**.** Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$. A correspondence (set-valued map) $F : X \rightrightarrows Y$ is *upper hemicontinuous* (uhc) at $x_0 \in X$ if for every open set $V \supseteq F(x_0)$, there exists an open set $U \ni x_0$ such that $F(x) \subseteq V$ for all $x \in U \cap X$.

Equivalently (when Y is compact), F is uhc at x_0 if whenever $x_k \to x_0$ and $y_k \in F(x_k)$ with $y_k \to y_0$, we have $y_0 \in F(x_0)$. This is the *closed graph* characterization.

Theorem 2.2 (Kakutani's Fixed Point Theorem)**.** *Let $K \subseteq \mathbb{R}^n$ be non-empty, compact, and convex. Let $F : K \rightrightarrows K$ be a correspondence such that:*

(i) $F(x)$ is non-empty for every $x \in K$.
(ii) $F(x)$ is convex for every $x \in K$.
(iii) F is upper hemicontinuous.

Then F has a fixed point: there exists $x^* \in K$ with $x^* \in F(x^*)$.

Proof. We prove via Brouwer's fixed point theorem. For each $k \in \mathbb{N}$, let T_k be a simplicial triangulation of K with mesh size at most $1/k$. For each vertex v of T_k, choose $f_k(v) \in F(v)$ (possible since $F(v)$ is non-empty). Extend f_k to all of K by barycentric interpolation: if $x = \sum_j \lambda_j v_j$ lies in a simplex with vertices $v_1, \ldots, v_p$, set $f_k(x) = \sum_j \lambda_j f_k(v_j)$.

Each $f_k : K \to K$ is continuous (linear on each simplex, consistent on shared faces by uniqueness of barycentric coordinates). Since K is convex and each $f_k(x)$ is a convex combination of points in K, we have $f_k(x) \in K$. By Brouwer's theorem, each f_k has a fixed point $x_k \in K$ with $f_k(x_k) = x_k$.

By compactness, (x_k) has a convergent subsequence $x_{k_l} \to x^*$. For each l, let $v_1^l, \ldots, v_p^l$ be the vertices of the simplex in T_{k_l} containing x_{k_l}, with barycentric coordinates λ_j^l. Then $x_{k_l} = \sum_j \lambda_j^l f_{k_l}(v_j^l)$ and $\|v_j^l - x_{k_l}\| \leq 1/k_l \to 0$, so $v_j^l \to x^*$.

Since $f_{k_l}(v_j^l) \in F(v_j^l) \subseteq K$ (compact), passing to a further subsequence (diagonal argument over finitely many vertex indices), we may assume $f_{k_l}(v_j^l) \to y_j$. By the closed-graph characterization of upper hemicontinuity: $v_j^l \to x^*$ and $f_{k_l}(v_j^l) \to y_j$ with $f_{k_l}(v_j^l) \in F(v_j^l)$ implies $y_j \in F(x^*)$.

The barycentric coordinates $\lambda_j^l \in [0,1]$ (compact), so passing to a further subsequence, $\lambda_j^l \to \lambda_j^*$ with $\sum_j \lambda_j^* = 1$, $\lambda_j^* \geq 0$. Taking the limit: $x^* = \sum_j \lambda_j^* y_j$. Since each $y_j \in F(x^*)$ and $F(x^*)$ is convex, we conclude $x^* \in F(x^*)$. □

2.2 Nash's Theorem

Key Result

Theorem 2.3 (Nash, 1950)**.** *Every finite normal-form game has at least one mixed strategy Nash equilibrium.*

Proof. Let $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a finite game with $|S_i| = m_i$. Define the set of mixed strategy profiles $\Delta = \prod_{i \in N} \Delta(S_i)$, which is compact and convex in $\mathbb{R}^{m_1 + \cdots + m_n}$.

Step 1: Define the best response correspondence. For each player i, define $\mathrm{BR}_i : \Delta_{-i} \rightrightarrows \Delta(S_i)$ by

$$\mathrm{BR}_i(\sigma_{-i}) = \arg\max_{\sigma_i \in \Delta(S_i)} u_i(\sigma_i, \sigma_{-i}).$$

Define the joint best response $\mathrm{BR} : \Delta \rightrightarrows \Delta$ by $\mathrm{BR}(\sigma) = \prod_{i \in N} \mathrm{BR}_i(\sigma_{-i})$.

Step 2: $\mathrm{BR}(\sigma)$ is non-empty. For each i, the map $\sigma_i \mapsto u_i(\sigma_i, \sigma_{-i})$ is continuous (linear) on the compact set $\Delta(S_i)$. By the extreme value theorem, the maximum is attained, so $\mathrm{BR}_i(\sigma_{-i}) \neq \emptyset$.

Step 3: $\mathrm{BR}(\sigma)$ is convex. Fix i and σ_{-i}. Let $\sigma_i, \sigma_i' \in \mathrm{BR}_i(\sigma_{-i})$ and $\lambda \in [0,1]$. Set $\hat{\sigma}_i = \lambda \sigma_i + (1-\lambda)\sigma_i'$. By linearity of u_i in σ_i:

$$u_i(\hat{\sigma}_i, \sigma_{-i}) = \lambda\, u_i(\sigma_i, \sigma_{-i}) + (1-\lambda)\, u_i(\sigma_i', \sigma_{-i}) = v_i^*,$$

where $v_i^* = \max_{\tau_i} u_i(\tau_i, \sigma_{-i})$. So $\hat{\sigma}_i \in \mathrm{BR}_i(\sigma_{-i})$, and $\mathrm{BR}_i(\sigma_{-i})$ is convex. Since $\mathrm{BR}(\sigma)$ is a product of convex sets, it is convex.

Step 4: BR is upper hemicontinuous. We use the closed-graph characterization. Let $\sigma^k \to \sigma^*$ in Δ and $\tau^k \in \mathrm{BR}(\sigma^k)$ with $\tau^k \to \tau^*$. Fix player i. Since $\tau_i^k \in \mathrm{BR}_i(\sigma_{-i}^k)$, for every

$\sigma_i \in \Delta(S_i)$:

$$u_i(\tau_i^k, \sigma_{-i}^k) \geq u_i(\sigma_i, \sigma_{-i}^k).$$

Since u_i is continuous on Δ, taking $k \to \infty$:

$$u_i(\tau_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*).$$

This holds for all σ_i, so $\tau_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$. Hence $\tau^* \in \mathrm{BR}(\sigma^*)$.

Step 5: Apply Kakutani. The correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$ satisfies all hypotheses of Theorem 2.2. Therefore there exists $\sigma^* \in \Delta$ with $\sigma^* \in \mathrm{BR}(\sigma^*)$, meaning $\sigma_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$ for all i, so σ^* is a Nash equilibrium. □

Remark 2.4 (Alternative Proof via Brouwer). An alternative proof avoids Kakutani by constructing a continuous function whose fixed points are NE. For each player i and pure strategy s_i^j, define the "excess payoff" $g_i^j(\sigma) = \max\{0, u_i(s_i^j, \sigma_{-i}) - u_i(\sigma)\}$, and set

$$f_i^j(\sigma) = \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)}.$$

One verifies: (i) f maps Δ to Δ; (ii) f is continuous; (iii) σ^* is a fixed point of f iff $g_i^j(\sigma^*) = 0$ for all i, j, iff σ^* is a NE. Brouwer's theorem yields a fixed point.

RL/ML Connection

Nash's theorem guarantees that every finite MARL problem (modeled as a stochastic game with finite state and action spaces) has at least one Nash equilibrium in the stage games. This justifies the goal of Nash-Q learning [Shoham and Leyton-Brown, 2008]. However, the theorem is non-constructive: computing a Nash equilibrium is PPAD-complete, meaning no polynomial-time algorithm is known.

3 Von Neumann's Minimax Theorem

3.1 Two-Player Zero-Sum Games

Definition 3.1 (Two-Player Zero-Sum Game)**.** A two-player game $(N, (S_1, S_2), (u_1, u_2))$ is *zero-sum* if $u_1(s) + u_2(s) = 0$ for all $s \in S$. It is specified by a single payoff matrix $A \in \mathbb{R}^{m \times n}$ where $u_1(i, j) = A_{ij}$ and $u_2(i, j) = -A_{ij}$. Player 1 (row) maximizes; player 2 (column) minimizes.

Definition 3.2 (Maximin and Minimax Values)**.** For a zero-sum game with payoff matrix A, define:

$$\underline{v} = \max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y \quad \text{(maximin value)},$$
$$\overline{v} = \min_{y \in \Delta_n} \max_{x \in \Delta_m} x^\top A y \quad \text{(minimax value)},$$

where Δ_k denotes the $(k-1)$-simplex in $\mathbb{R}^k$.

Lemma 3.3 (Maximin $\leq$ Minimax)**.** *For any matrix $A \in \mathbb{R}^{m \times n}$, $\underline{v} \leq \overline{v}$.*

Proof. For any fixed $x \in \Delta_m$ and $y \in \Delta_n$:

$$\min_{y' \in \Delta_n} x^\top A y' \leq x^\top A y \leq \max_{x' \in \Delta_m} (x')^\top A y.$$

Taking max over x on the left side and min over y on the right:

$$\underline{v} = \max_x \min_{y'} x^\top A y' \leq \min_y \max_{x'} (x')^\top A y = \overline{v}.$$ □

Key Result

Theorem 3.4 (Von Neumann, 1928). *For every matrix $A \in \mathbb{R}^{m\times n}$,*

$$\max_{x\in\Delta_m} \min_{y\in\Delta_n} x^\top Ay = \min_{y\in\Delta_n} \max_{x\in\Delta_m} x^\top Ay.$$

The common value $v^* = \underline{v} = \overline{v}$ is the value *of the game.*

Proof. We prove $\underline{v} \geq \overline{v}$ using LP duality; combined with Lemma 3.3, equality follows.

Step 1: Reformulate as linear programs. Since $y \mapsto x^\top Ay$ is linear in y and Δ_n is a simplex, the minimum over y is attained at a vertex e_j, giving $\min_y x^\top Ay = \min_{j=1,\dots,n}(A^\top x)_j$. Thus $\underline{v} = \max_{x\in\Delta_m} \min_j (A^\top x)_j$. Introducing an auxiliary variable v:

$$\begin{aligned} \text{(P)} \quad & \max \ v \\ & \text{s.t. } A^\top x \geq v\,\mathbf{1}_n, \quad x \in \Delta_m, \quad v \in \mathbb{R}. \end{aligned} \tag{1}$$

Step 2: The dual LP. Similarly, $\overline{v} = \min_{y\in\Delta_n} \max_i (Ay)_i$, which gives:

$$\begin{aligned} \text{(D)} \quad & \min \ w \\ & \text{s.t. } Ay \leq w\,\mathbf{1}_m, \quad y \in \Delta_n, \quad w \in \mathbb{R}. \end{aligned} \tag{2}$$

Step 3: Apply strong LP duality. Programs (P) and (D) are LP duals of each other. Both are feasible: take $x = (1/m, \dots, 1/m)^\top$ for (P) and $y = (1/n, \dots, 1/n)^\top$ for (D). Both are bounded (the feasible sets are compact). By strong LP duality, the optimal values coincide:

$$\underline{v} = \text{opt(P)} = \text{opt(D)} = \overline{v}. \qquad \square$$

Corollary 3.5 (NE in Zero-Sum Games). *In a two-player zero-sum game with payoff matrix A and value v^*:*

(a) (x^*, y^*) is a NE iff x^* is maximin and y^* is minimax.
(b) All NE yield payoffs $u_1 = v^*$, $u_2 = -v^*$.
(c) NE strategies are interchangeable: if (x^*, y^*) and (x^{**}, y^{**}) are both NE, then so are (x^*, y^{**}) and (x^{**}, y^*).

Proof. **(a)** **($\Rightarrow$):** Let (x^*, y^*) be a NE. Then x^* best responds to y^*: $(x^*)^\top Ay^* = \max_x x^\top Ay^*$. Also y^* best responds to x^*: $(x^*)^\top Ay^* = \min_y (x^*)^\top Ay$. Therefore $\min_y (x^*)^\top Ay \leq \underline{v} \leq \overline{v} \leq \max_x x^\top Ay^*$, and the minimax theorem gives $\underline{v} = \overline{v} = v^*$, so all inequalities are equalities: x^* achieves the maximin and y^* achieves the minimax.

($\Leftarrow$): If x^* achieves the maximin and y^* achieves the minimax, then $(x^*)^\top Ay^* \geq \min_y (x^*)^\top Ay = v^* = \max_x x^\top Ay^* \geq (x^*)^\top Ay^*$, forcing equality. For any x: $x^\top Ay^* \leq v^* = (x^*)^\top Ay^*$, so x^* best responds to y^*. Similarly y^* best responds to x^*. Hence (x^*, y^*) is a NE.

(b) Immediate from part (a).

(c) If x^* and x^{**} are both maximin and y^*, y^{**} both minimax, then any maximin-minimax pair forms a NE by part (a). $\square$

3.2 Solving Zero-Sum Games via LP

Proposition 3.6 (Zero-Sum Games via LP). *The maximin strategy x^* and game value v^* are computed by solving LP (1). The minimax strategy y^* is the dual solution from LP (2). Both LPs have polynomial-time algorithms, so computing NE in two-player zero-sum games is in* P.

Proof. This follows from the reformulation in the proof of Theorem 3.4. LP (1) has $m + 1$ variables and $n + m + 1$ constraints, solvable in polynomial time by interior point methods. $\square$

Warning

While NE in zero-sum games is polynomial via LP, computing a NE in a general two-player game is PPAD-complete. This complexity barrier is a fundamental challenge for MARL.

RL/ML Connection

The minimax theorem is the foundation of: (i) **Minimax-Q learning** [Shoham and Leyton-Brown, 2008], where the Bellman operator computes $V(s) = \max_x \min_y x^\top Q(s,\cdot,\cdot)y$; (ii) **AlphaZero**, where MCTS approximates minimax values; (iii) **GANs**, where $\min_G \max_D V(D,G)$ is a continuous zero-sum game whose saddle points are justified by minimax theory.

4 Correlated Equilibrium

Definition 4.1 (Correlated Equilibrium)**.** A *correlated equilibrium* (CE) of a finite game $\Gamma = (N,(S_i),(u_i))$ is a probability distribution π over $S = \prod_{i\in N} S_i$ such that for every player i and every pair $s_i, s_i' \in S_i$:

$$\sum_{s_{-i}\in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i, s_{-i}) \geq \sum_{s_{-i}\in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i', s_{-i}).$$

Equivalently, if a mediator draws s from π and privately recommends s_i to player i, no player has incentive to deviate.

Key Result

Theorem 4.2 (Every NE Is a CE)**.** *If $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed strategy NE, then the product distribution $\pi(s) = \prod_{i\in N} \sigma_i^*(s_i)$ is a correlated equilibrium.*

Proof. Fix player i and strategies $s_i, s_i' \in S_i$. If $\sigma_i^*(s_i) = 0$, both sides of the CE constraint are 0 and the inequality holds trivially. If $\sigma_i^*(s_i) > 0$, since π is a product distribution, $\pi(s_i, s_{-i}) = \sigma_i^*(s_i) \prod_{j\neq i} \sigma_j^*(s_j)$. Dividing both sides of the CE constraint by $\sigma_i^*(s_i) > 0$ gives:

$$\sum_{s_{-i}} \left(\prod_{j\neq i} \sigma_j^*(s_j) \right) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}} \left(\prod_{j\neq i} \sigma_j^*(s_j) \right) u_i(s_i', s_{-i}),$$

i.e., $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$. Since $s_i \in \operatorname{supp}(\sigma_i^*)$, the indifference principle (Theorem 1.6) gives exactly this inequality. □

Proposition 4.3 (CE as a Linear Program)**.** *The set of correlated equilibria is a convex polytope defined by the linear CE constraints, $\pi(s) \geq 0$, and $\sum_s \pi(s) = 1$. Optimizing any linear objective over the CE set is an LP.*

Proof. The CE constraints are linear in $\{\pi(s)\}_{s\in S}$:

$$\sum_{s_{-i}} \pi(s_i, s_{-i}) \big[u_i(s_i, s_{-i}) - u_i(s_i', s_{-i}) \big] \geq 0 \quad \text{for all } i, s_i, s_i'.$$

Combined with the simplex constraints, the feasible set is a polytope. □

Example 4.4 (CE in Battle of the Sexes)**.** Consider the "traffic light" CE: a mediator recommends (O, O) with probability 1/2 and (F, F) with probability 1/2. Then $\pi(O, O) = \pi(F, F) = 1/2$ and $\pi(O, F) = \pi(F, O) = 0$.

Check for player 1 given recommendation O: $\pi(O, O)\, u_1(O, O) = 3/2 \geq \pi(O, O)\, u_1(F, O) = 0$. Given F: $\pi(F, F)\, u_1(F, F) = 1/2 \geq \pi(F, F)\, u_1(O, F) = 0$. Similarly for player 2. Expected payoffs: 2 for each player, Pareto-dominating the mixed NE payoffs of 3/4 each.

RL/ML Connection

Correlated equilibrium has key advantages for MARL: (i) finding a CE is an LP (polynomial time), unlike NE (PPAD-complete); (ii) no-regret learning dynamics (e.g., regret matching) converge to CE naturally; (iii) CE can achieve better social welfare than any NE [Aumann, 1974].

5 Exercises

Exercise 5.1 (Mixed NE Computation)**.** Compute all Nash equilibria (pure and mixed) of the game:

	L	R
T	$(4, 2)$	$(1, 3)$
B	$(2, 3)$	$(3, 1)$

Verify the indifference principle for each mixed NE.

Exercise 5.2 (Support Enumeration)**.** Consider the 3×3 game:

	L	C	R
T	$(1, 1)$	$(0, 0)$	$(0, 0)$
M	$(0, 0)$	$(1, 1)$	$(0, 0)$
B	$(0, 0)$	$(0, 0)$	$(1, 1)$

(a) Find all pure strategy NE.
(b) Find all mixed NE with support of size 2 for each player.
(c) Find the fully mixed NE (support of size 3 for each player).
(d) Show that the set of NE is not convex.

Exercise 5.3 (Minimax Computation)**.** For the zero-sum game with payoff matrix

$$A = \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix},$$

compute the value v^* and optimal mixed strategies for both players. Formulate as an LP and verify the minimax theorem directly.

Exercise 5.4 (Correlated Equilibrium Construction)**.** In the game of Chicken:

	D	S
D	$(-5, -5)$	$(2, -1)$
S	$(-1, 2)$	$(0, 0)$

(a) Find all pure and mixed Nash equilibria.
(b) Construct a CE that Pareto-dominates the mixed NE.
(c) Write the CE incentive constraints explicitly and verify your solution satisfies them.

Exercise 5.5 (CE Polytope Structure)**.** Prove that the set of correlated equilibria is closed under convex combinations: if π and π' are CE and $\lambda \in [0,1]$, then $\lambda\pi + (1-\lambda)\pi'$ is a CE. Conclude that the CE set is a convex polytope.

Exercise 5.6 (Kakutani Conditions)**.** Consider the two-player game where $S_1 = S_2 = \{A, B\}$ with payoffs $(3,3)$ at (A,A), $(0,0)$ at (A,B) and (B,A), and $(1,1)$ at (B,B).

(a) Write the best response correspondence $\mathrm{BR}_i(\sigma_{-i})$ explicitly as a function of the opponent's mixing probability.
(b) Verify that BR_i is non-empty-valued, convex-valued, and upper hemicontinuous.
(c) Find all fixed points of BR (i.e., all NE).

Exercise 5.7 (Minimax for 3×3)**.** For the zero-sum game with payoff matrix

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

show that the game value is $v^* = 0$ and find all maximin and minimax strategies. *Hint:* consider the role of the third row/column.

Exercise 5.8 (NE $\subsetneq$ CE)**.** Give an example of a 2×2 game and a correlated equilibrium π that is *not* a product distribution (and hence not a NE viewed as a joint distribution). Prove that your π satisfies the CE constraints but cannot be written as $\pi(s) = \sigma_1(s_1)\,\sigma_2(s_2)$ for any mixed strategy profile (σ_1, σ_2).

Summary. This lesson established: (1) mixed strategies and the multilinear expected payoff; (2) the indifference principle characterizing mixed NE support; (3) Nash's existence theorem via Kakutani's fixed point theorem; (4) the minimax theorem for zero-sum games via LP duality, with the corollary that zero-sum NE are polynomial to compute; and (5) correlated equilibrium as a computationally tractable relaxation of NE. These results form the strategic foundation for multi-agent RL in Phase 05.

References

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